PREP 2, LESSONS 3–4

From plum pudding to the nucleus

The plum pudding model; alpha scattering; the nuclear model; energy levels

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Describe the structure of an atom. DN 224 [1]

A positive nucleus of protons and neutrons, surrounded by negative electrons.

2 Define isotopes. DN 242 [1]

Atoms of the same element with different numbers of neutrons.

3 State how to calculate the number of neutrons in an atom. DN 716 [1]

Number of neutrons = mass number − atomic number

Key ideas LESSONS 3–4

Show your working

4 Describe the plum pudding model of the atom. [1]

A ball of positive charge with negative electrons embedded in it ✓

5 In the alpha scattering experiment, alpha particles were fired at thin gold foil.

a Most alpha particles passed straight through. What does this show? [1]

Most of the atom is empty space ✓

b A very few bounced back. Explain what this shows about the nucleus. [2]

The mass is concentrated in a tiny nucleus ✓

which is positively charged, so it repels the positive alpha particles ✓

6 In the Bohr model, electrons orbit the nucleus at [any distance / specific distances] [1]

7 An electron moves to a higher energy level when the atom [absorbs / emits] electromagnetic radiation. [1]

8 The evidence for the neutron came from [Niels Bohr / James Chadwick / Ernest Rutherford] [1]

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